AI ASSISTED CODING

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Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation

Task 1 – Portfolio Website Design

You are building a personal portfolio website to showcase your

work.

Requirements:

• Create sections for About Me, Projects, and Contact.

• Use AI to:

o Suggest color palettes and typography.

o Create a responsive layout with Grid/Flexbox.

o Add smooth scrolling navigation.

Lab_14.1.html > html

```html
#Task - 1
#Create a responsive personal portfolio website using HTML, CSS, and JavaScript. The website should include three sections: About
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Personal Portfolio</title>
    <style>
        /* Modern color palette */
        :root {
            --primary-color: #4A90E2;
            --secondary-color: #50E3C2;
            --background-color: #F5F5F5;
            --text-color: #333333;
            --accent-color: #D0021B;
        }

        /* Global styles */
        body {
            margin: 0;
            font-family: 'Arial', sans-serif;
            background-color: var(--background-color);
            color: var(--text-color);
        }

        /* Navigation bar */
        nav {
            position: fixed;
            top: 0;
            width: 100%;
            background-color: var(--primary-color);
            display: flex;
            justify-content: center;
            padding: 1rem 0;
            z-index: 1000;
        }
```

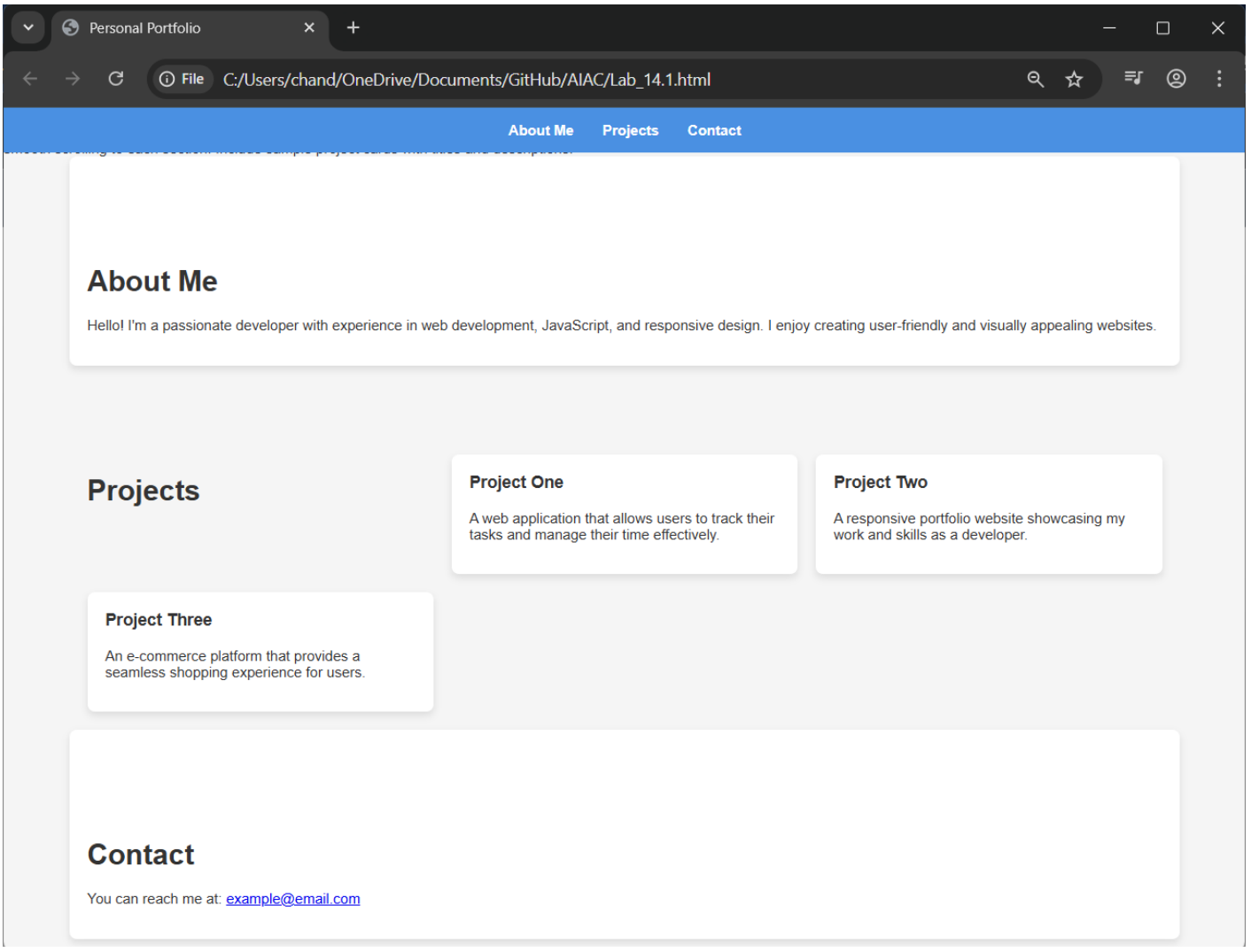
Lab_14.1.html > html

```html
<html lang="en">
<body>
    <nav>
        <a href="#contact">Contact</a>
    </nav>

    <section id="about">
        <h1>About Me</h1>
        <p>Hello! I'm a passionate developer with experience in web development, JavaScript, and responsive design. I enjoy creat
    </section>

    <section id="projects">
        <h1>Projects</h1>
        <div class="project-card">
            <h3>Project One</h3>
            <p>A web application that allows users to track their tasks and manage their time effectively.</p>
        </div>
        <div class="project-card">
            <h3>Project Two</h3>
            <p>A responsive portfolio website showcasing my work and skills as a developer.</p>
        </div>
        <div class="project-card">
            <h3>Project Three</h3>
            <p>An e-commerce platform that provides a seamless shopping experience for users.</p>
        </div>
    </section>

    <section id="contact">
        <h1>Contact</h1>
        <p>You can reach me at:
            <a href="mailto:example@email.com">example@email.com</a>
        </p>
    </section>
</body>
</html>
```



Task 2 – Online Store Product Page

Design a product display page for an online store.

Requirements:

• Display product image, title, price, and "Add to Cart" button.

• Use AI to:

o Style with BEM methodology.

o Make layout responsive.

o Add hover effects and "Add to Cart" alert.

Lab_14.1 > Lab_14.1_2.html > html > head > style > .product-card__button:hover

```html
#Task - 2
#Create a responsive product page for an online store using HTML, CSS, and JavaScript. Display a product image, product title, price, description, and an 'Add to Cart' button. Style the page using the BEM (Block Element Modifier) CSS methodology. Make the layout responsive for mobile and desktop. Add hover effects to the product card and button. When the user clicks 'Add to Cart', show a JavaScript alert confirming the product was added to the cart.
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Product Page</title>
    <style>
        /* BEM CSS Methodology */
        .product-card {
            background-color: #fff;
            border-radius: 8px;
            box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
            max-width: 400px;
            margin: auto;
            padding: 20px;
            text-align: center;
        }

        .product-card__image {
            width: 100%;
            border-radius: 8px;
        }

        .product-card__title {
            font-size: 1.5rem;
            margin: 15px 0;
        }

        .product-card__price {
            font-size: 1.25rem;
            color: #4A90E2;
            margin-bottom: 15px;
```

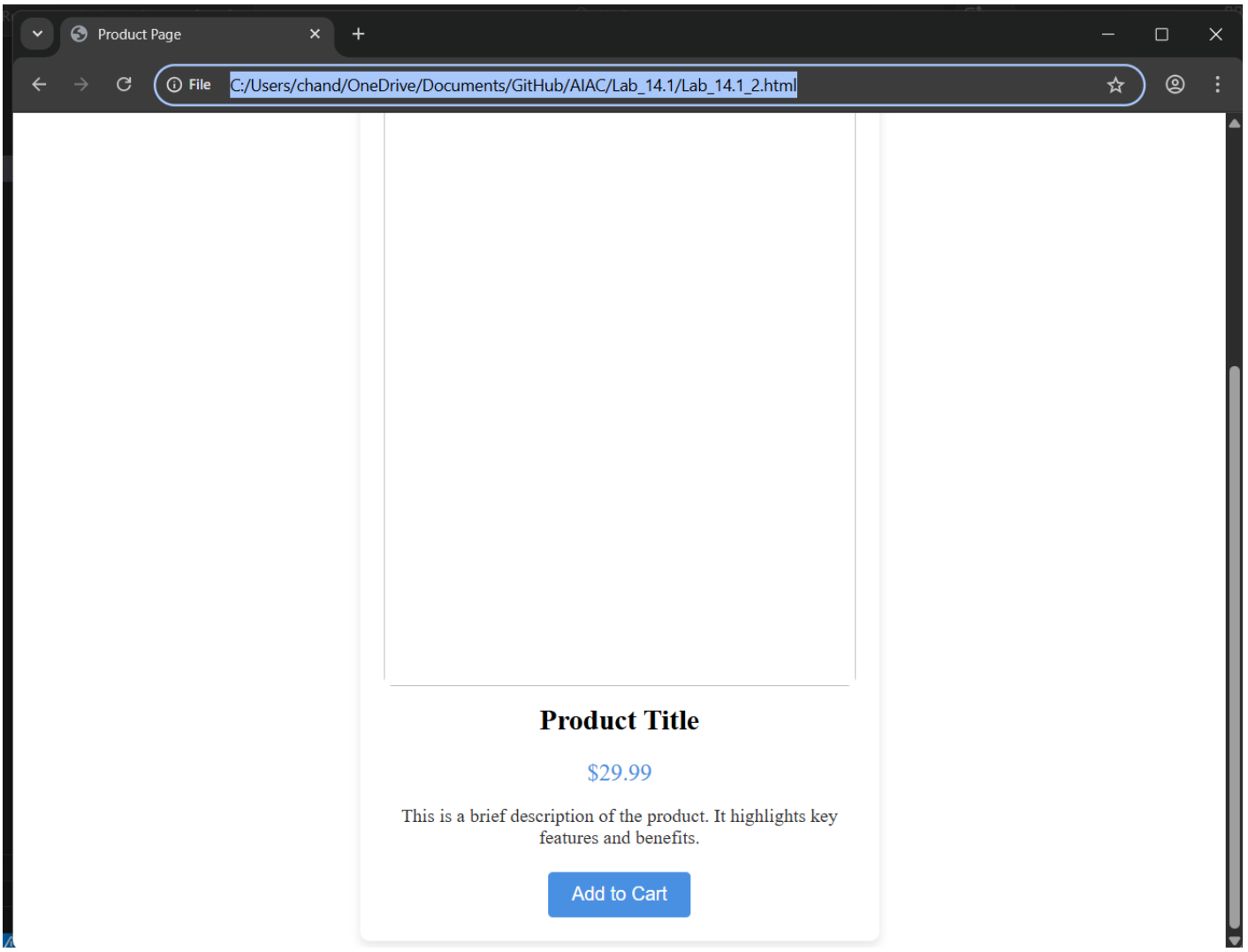
Lab_14.1 > Lab_14.1_2.html > html > head > style > .product-card__button:hover

```html
<html lang="en">
<head>
    <style>
        .product-card__button {
        }

        .product-card__button:hover {
            background-color: #D0021B;
        }

        /* Responsive layout */
        @media (max-width: 600px) {
            .product-card {
                max-width: 100%;
                padding: 15px;
            }
        }
    </style>
</head>
<body>
    <div class="product-card">
        <img src="product-image.jpg" alt="Product Image" class="product-card__image">
        <h2 class="product-card__title">Product Title</h2>
        <p class="product-card__price">$29.99</p>
        <p class="product-card__description">This is a brief description of the product. It highlights key features and benefits.<
        <button class="product-card__button" onclick="addToCart()">Add to Cart</button>
    </div>

    <script>
        function addToCart() {
            alert('Product added to cart!');
        }
    </script>
</body>
</html>
```



Task 3 – Event Registration Form

Build an event registration form for a conference.

Requirements:

• Collect name, email, phone number, and session selection.

• Use AI to:

o Add form validation with JavaScript.

o Make the form accessible with labels and ARIA.

o Style with a professional look.

Lab_14.1 > Lab_14.1_3.html > html > head > style > body

```html
#Task - 3
#Create an event registration form using HTML, CSS, and JavaScript. The form should collect name, email, phone number, and
session selection using a dropdown menu. Add JavaScript form validation to check required fields, email format, and phone number
 length. Ensure accessibility by using proper labels and ARIA attributes. Style the form with a clean and professional design
 suitable for a conference website. Make the form responsive.
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Event Registration</title>
    <style>
        /* Clean and professional design */
        body {
            font-family: 'Arial', sans-serif;
            background-color: #f5f5f5;
            color: #333;
            margin: 0;
            padding: 20px;
        }

        .form-container {
            max-width: 600px;
            margin: auto;
            background-color: #fff;
            padding: 20px;
            border-radius: 8px;
            box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
        }

        label {
            display: block;
            margin-bottom: 5px;
            font-weight: bold;
        }

        input, select {
```

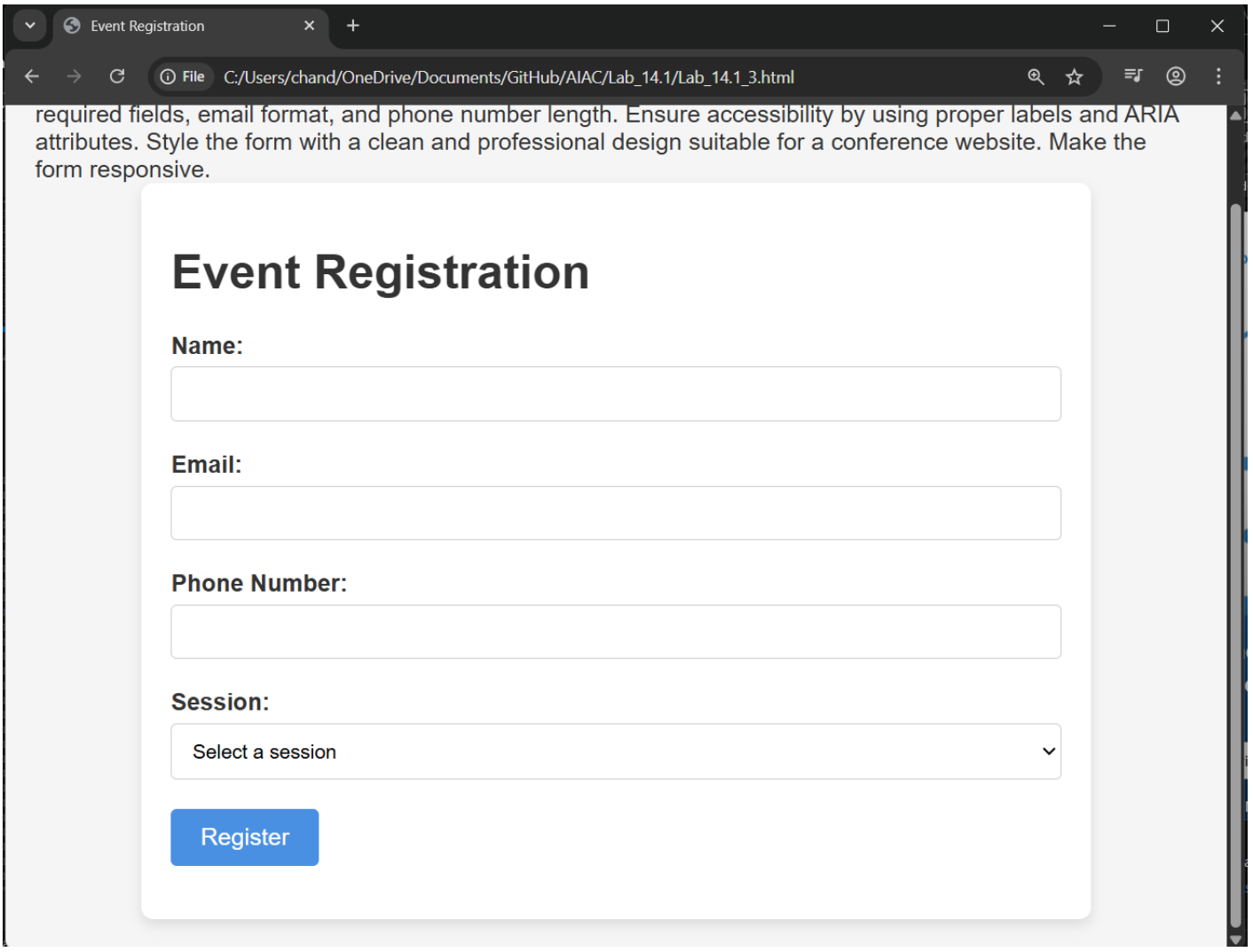
Lab_14.1 > Lab_14.1_3.html > html > head > style > body

```html
<html lang="en">
<body>
    <script>
        document.getElementById('registrationForm').addEventListener('submit', function(event) {
            event.preventDefault();

            // Get form values
            const name = document.getElementById('name').value.trim();
            const email = document.getElementById('email').value.trim();
            const phone = document.getElementById('phone').value.trim();
            const session = document.getElementById('session').value;

            // Simple validation
            if (!name || !email || !phone || !session) {
                alert('Please fill in all required fields.');
                return;
            }

            // Email format validation
            const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
            if (!emailRegex.test(email)) {
                alert('Please enter a valid email address.');
                return;
            }
            // Phone number length validation (assuming 10 digits)
            const phoneDigits = phone.replace(/\D/g, ''); // Remove non-digit characters
            if (phoneDigits.length !== 10) {
                alert('Please enter a valid phone number with 10 digits.');
                return;
            }
            // If validation passes, submit the form (you can replace this with an actual submission)
            alert('Registration successful!');
        });
    </script>
</body>
</html>
```



Task 4 – Resume Website

Build a one-page online resume.

Requirements:

• Sections: Profile Summary, Skills, Education, Experience.

• Use AI to:

o Propose professional typography styles.

o Align sections using CSS Grid.

o Add smooth scroll navigation between sections.

Lab_14.1 > Lab_14.1_4.html > ...

```html
#Task - 4
#Create a one-page resume with Profile, Skills, Education, and Experience. Use CSS Grid, professional fonts,
 and smooth scrolling navigation.
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>One-Page Resume</title>
    <style>
        /* Professional fonts */
        @import url('https://fonts.googleapis.com/css2?family=Roboto:wght@400;700&display=swap');

        body {
            font-family: 'Roboto', sans-serif;
            background-color: #f5f5f5;
            color: #333;
            margin: 0;
            padding: 20px;
        }

        .resume-container {
            max-width: 800px;
            margin: auto;
            background-color: #fff;
            padding: 20px;
            border-radius: 8px;
            box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
        }

        h1, h2 {
            color: #4A90E2;
        }

        /* CSS Grid layout */
        .grid-container {
            display: grid;
```

Lab_14.1 > Lab_14.1_4.html > ...

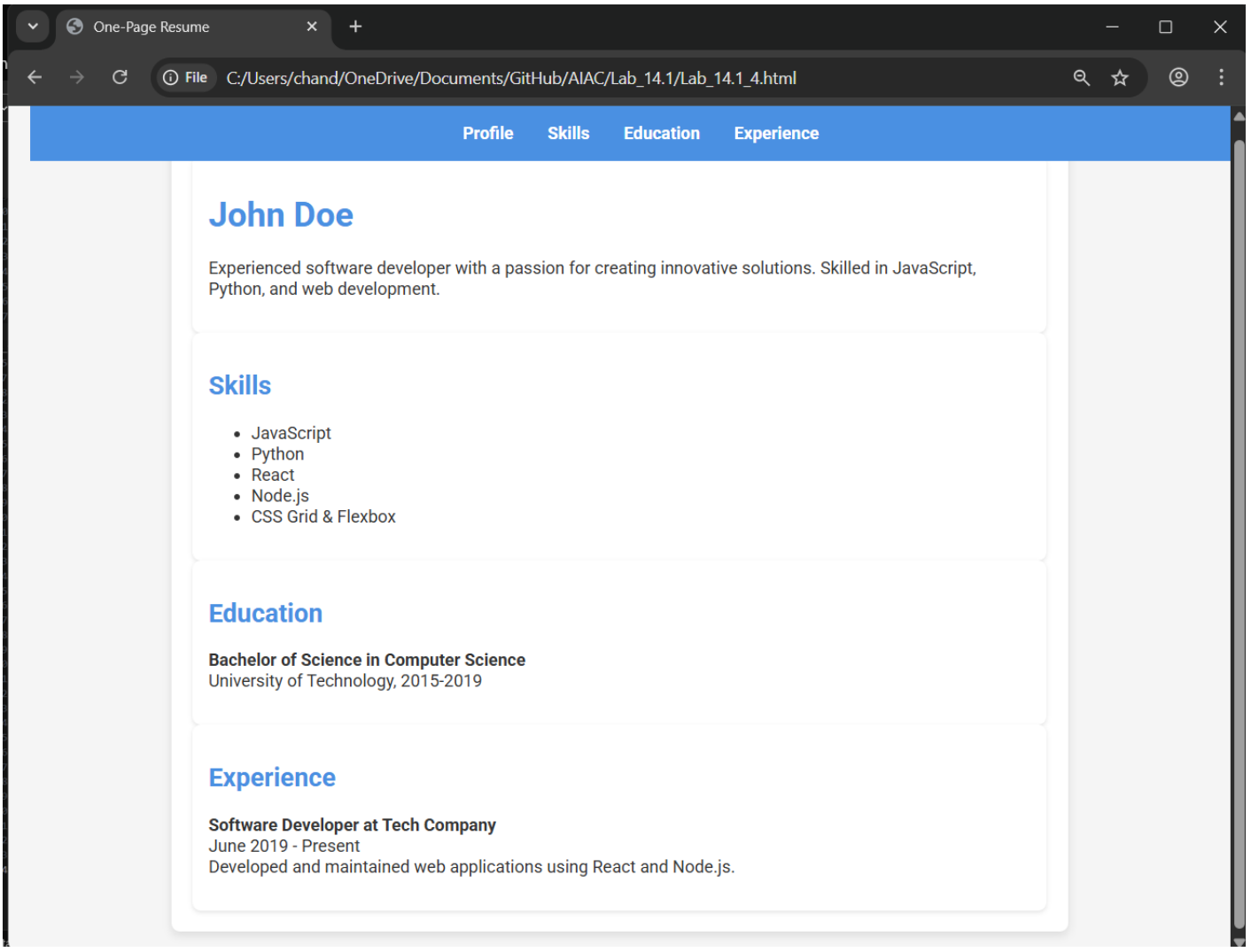
```html
<html lang="en">
<body>
    <nav>
        <a href="#experience">Experience</a>
    </nav>

    <div class="resume-container">
        <section id="profile" class="section">
            <h1>John Doe</h1>
            <p>Experienced software developer with a passion for creating innovative solutions. Skilled in JavaScript, Python, and
        </section>

        <section id="skills" class="section">
            <h2>Skills</h2>
            <ul>
                <li>JavaScript</li>
                <li>Python</li>
                <li>React</li>
                <li>Node.js</li>
                <li>CSS Grid & Flexbox</li>
            </ul>
        </section>

        <section id="education" class="section">
            <h2>Education</h2>
            <p><strong>Bachelor of Science in Computer Science</strong><br>University of Technology, 2015-2019</p>
        </section>

        <section id="experience" class="section">
            <h2>Experience</h2>
            <p><strong>Software Developer at Tech Company</strong><br>June 2019 - Present<br>Developed and maintained web applicat
        </section>
    </div>
</body>
</html>
```



Task 5 – Restaurant Menu Page

Design a digital menu page for a restaurant.

Requirements:

• Display menu items with name, description, and price.

• Use AI to:

o Suggest a food-friendly color palette.

o Design a responsive card-based layout.
o Add hover effects on menu items.

Lab_14.1 > Lab_14.1_5.html > ...

```html
#Task - 5
#Create a responsive restaurant menu page with item name, description, and price. Use card layout,
 food-style colors, and hover effects.
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Restaurant Menu</title>
    <style>
        /* Food-style colors */
        body {
            font-family: 'Arial', sans-serif;
            background-color: #fff8e1;
            color: #333;
            margin: 0;
            padding: 20px;
        }

        .menu-container {
            max-width: 1200px;
            margin: auto;
            display: grid;
            grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
            gap: 20px;
        }

        .menu-item {
            background-color: #fff3e0;
            border-radius: 8px;
            box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
            padding: 20px;
            text-align: center;
            transition: transform 0.3s ease, box-shadow 0.3s ease;
        }

        .menu-item:hover {
```

Restaurant Menu

File C:/Users/chand/OneDrive/Documents/GitHub/AIAC/Lab_14.1/Lab_14.1_5.html

#Task - 5 #Create a responsive restaurant menu page with item name, description, and price. Use card layout, food-style colors, and hover effects.

Margherita Pizza

A classic pizza with tomato sauce, mozzarella, and basil.

$12.99

Caesar Salad

Fresh romaine lettuce with Caesar dressing, croutons, and parmesan cheese.

$8.99